

CMA: Identity-Conditioned Relational Segmentation with Supplied Localized Entity Memory

Working manuscript shell

Abstract

Segmenting the correct helmet among visually similar workers requires resolving whose equipment a query refers to. We study relational segmentation conditioned on supplied identity-localized entity memory: an appearance crop, miner mask and bounding box. CMA injects crop and box features at the language model’s REF input and combines REF hidden state with mask and box geometry in the SAM prompt, alongside a paired counterfactual identity-ranking training objective. Evaluation fixes the observation and query while switching the supplied miner identity. Reference appearance comes from the same condition image; supplied geometry remains fixed. Under a fixed photometric/noise/blur stressor, the primary documented-protocol-disjoint confirmation yields 58% versus 2% CMSA and 84% versus 46% Memory Fidelity for CMA versus pinned SegLLM through their native memory interfaces. This system-level result supports supplied-memory use, not autonomous tracking or memory acquisition. Targets are reconstructed pseudo labels, exact historical training exposure remains unknown, and the comparison does not establish isolated component gains or lower degradation sensitivity.

1 Introduction

In scenes containing visually similar workers, locating a helmet is only part of the task: a relational request must identify whose helmet is intended. Small equipment regions provide limited visual evidence, and darkness, dust, blur and glare can further obscure that evidence. A category-correct mask can therefore answer the wrong worker-specific question. We study this ambiguity through supplied miner identity memory: a reference identifies the worker, while a shared query requests the helmet belonging to that worker.

An aggregate segmentation score does not by itself test whether this reference controls target identity. A model can repeatedly select a plausible helmet without following a change in the requested worker. We consequently hold the observation and relational query fixed and switch only the supplied identity memory. Each switch has a corresponding helmet target. This controlled intervention distinguishes memory-dependent target selection from a preference for one visually salient object, while a localization criterion prevents identity ranking alone from being

counted as successful segmentation.

Our implemented CMA design conditions segmentation on entity memory at two stages. Reference appearance and bounding-box geometry are mapped into the language-model input at the [REF] token. The associated language representation and reference-mask geometry then condition the SAM prompt used to decode the equipment mask. A paired counterfactual objective encourages a memory-conditioned helmet prediction to overlap its own target more than the other worker’s target. These mechanisms describe the implemented system; our comparison does not isolate the gain of each component.

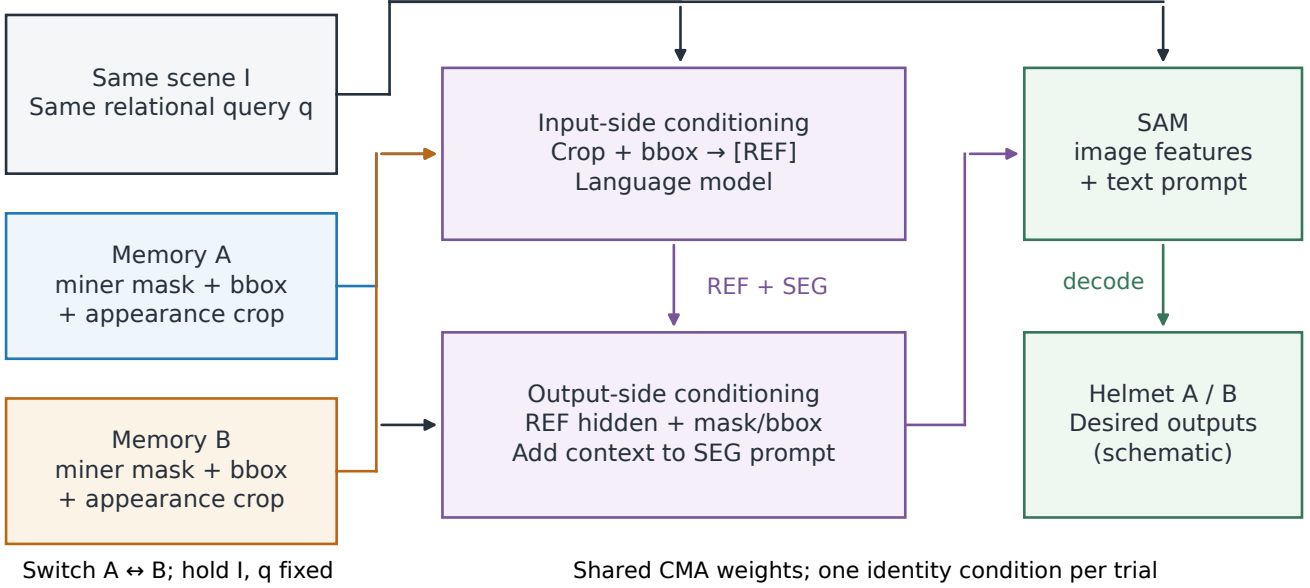
We summarize the contributions as follows:

- We formulate counterfactual identity-memory relational perception for visually similar workers and small PPE targets, asking whether a supplied worker identity controls which equipment instance is segmented.
- We instantiate dual-stage REF conditioning through appearance/bbox input injection and reference-conditioned segmentation prompts, with a paired counterfactual identity-ranking objective. We evaluate supplied-memory use by switching identity under a fixed observation/query and measuring CMSA, Memory Fidelity, Identity Error Rate and mIoU, without attributing isolated causal gains to components that lack matched ablations.
- We provide a frozen system-level comparison under a fixed coal-mine-style compound degradation. The primary documented-protocol-disjoint confirmation shows stronger absolute degraded segmentation and identity fidelity than the pinned SegLLM comparator. Exact historical exposure remains unknown; this is not a claim of lower degradation sensitivity or exact training-unseen generalization.

Validated input and stressor. Throughout the experiments, memory is a supplied identity-localized entity state: a miner mask, bounding box and appearance crop from the same condition image. Worker geometry is supplied and fixed. Dust and glare above describe broader motivation only; the tested `target15_b` stressor combines gamma darkening, contrast/scale reduction, Gaussian noise and Gaussian blur. It does not simulate dust, glare or occlusion.

Supplied identity-localized entity memory

Same-condition appearance; externally supplied, fixed miner geometry
image I



Training only: paired identity rank

$$S_{ij} = \text{softIoU}(p_i, Y_j)$$

$$[\mu + \max_{j \neq i} S_{ij} - S_{ij}]_+$$

Alongside text, mask and REF reconstruction losses

Offline evaluation: saved binary masks

JAA	JAB
JBA	JBB

Fidelity: correct > wrong

CMSA: both correct + IoU ≥ 0.5

IER: stronger wrong IoU ≥ 0.5

Figure 1: CMA conditioned on supplied identity-localized entity memory. Each alternative A/B contains a miner mask, bounding box and appearance crop from the same condition image. Geometry is supplied and fixed. Crop/bbox features condition the language model at REF; REF hidden state and mask/bbox geometry augment the SEG-derived SAM prompt. Shared weights decode the relational helmet target. A/B outputs are schematic desired identities, not additional measured predictions. The separate training inset shows soft-IoU ranking; the offline evaluation inset uses binary IoU. Neither inset supplies helmet targets to inference.

2 Method

2.1 Task and supplied identity memory

Let I be an observation, q a relational query, R_i a supplied miner reference mask, b_i its normalized bounding box, and Y_i the corresponding helmet target. We denote the supplied identity condition by $M_i = (R_i, b_i, C(I, R_i, b_i))$, where C masks the background, crops the reference box, zero-pads to a square and applies the reference-image processor. For two workers in the same image, the desired behavior is

$$f(I, q, M_A) \rightarrow Y_A, \quad f(I, q, M_B) \rightarrow Y_B. \quad (1)$$

Here memory denotes supplied identity evidence, not an autonomously maintained state. Under degradation the reference appearance is taken from the same degraded ob-

servation as the main image; the supplied mask and box identify the worker. This protocol tests memory use under degraded appearance, not recovery of an independently corrupted reference mask.

2.2 Dual-stage identity-memory conditioning

The following equations abstract the verified implementation rather than specifying every layer. First, visual tokens from the reference crop are averaged and combined with a bounding-box projection:

$$v_i = \text{mean } E_{\text{vis}}(C(I, R_i, b_i)), \quad (2)$$

$$r_i = P_{\text{in}}(v_i + B(b_i)), \quad (3)$$

$$e'_{\text{REF},i} = e_{\text{REF}} + \alpha r_i. \quad (4)$$

The frozen configuration uses $\alpha = 0.5$. This scalar addition is implemented as `gated_add`; it is not an estimated memory-reliability gate. The language model processes the image and conversation with the injected reference embedding.

For output-side conditioning, let h_i denote the projected REF-associated language hidden representation and z_i the SEG-derived text prompt. The supplied mask is bilinearly resized to 16×16 , flattened and concatenated with b_i before a geometry projection G . The memory prompt is

$$g_i = G([\text{vec}(\text{resize}_{16}(R_i)); b_i]), \quad (5)$$

$$c_i = s \text{ cap}_{20}(h_i + g_i), \quad (6)$$

$$t_i = \text{clip}_{[-50, 50]}(z_i + c_i), \quad (7)$$

$$\hat{Y}_i = D_{\text{SAM}}(E_{\text{SAM}}(I), t_i). \quad (8)$$

The operator cap_{20} bounds vector norm; s is a learned reference scale, whose configured initialization is 0.2 rather than its final learned value. Validity flags restrict reference conditioning to the relevant rows. The frozen prompt mode adds context to the text prompt before SAM prompt encoding and single-mask decoding. Its contribution is evaluated as part of the complete system, not through a component-isolated ablation.

2.3 Counterfactual identity objective

The documented training construction groups worker-helmet pairs from the same image and assigns the same second-round query to each pair. Training outputs alternate miner and helmet masks. In addition to token cross-entropy and weighted mask BCE/Dice losses, an auxiliary REF decoding branch reconstructs the supplied miner mask on valid reference rows. Writing these normalized losses as $\mathcal{L}_{\text{text}}$, $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{ref}}$, the implemented objective has the form

$$\mathcal{L} = \mathcal{L}_{\text{text}} + \mathcal{L}_{\text{seg}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (9)$$

For a grouped example, let p_i be the sigmoid helmet prediction conditioned on identity i . The differentiable overlap matrix and ranking term are

$$S_{ij} = \frac{\sum_x p_i(x) Y_j(x)}{\max\{\sum_x [p_i(x) + Y_j(x) - p_i(x) Y_j(x)], \epsilon\}}, \quad (10)$$

$$\mathcal{L}_{\text{rank}} = \frac{1}{N} \sum_i \left[\mu + \max_{j \neq i} S_{ij} - S_{ii} \right]_+. \quad (11)$$

Here N is the number of helmet references in the group. The code uses $\epsilon = 10^{-6}$ and selects helmet entries from the alternating mask sequence; the rank term is applied only to eligible groups with at least two helmet targets. The saved w15 configuration records $\lambda_{\text{ref}} = 1.0$, $\lambda_{\text{rank}} = 1.5$ and $\mu = 0.05$. These are hash-verified checkpoint metadata. Exact historical run-bound data and CE/BCE/Dice coefficient overrides remain unknown, so launcher defaults are not presented as verified historical settings. We make no claim that the rank term alone caused the whole-system comparison gap.

2.4 Counterfactual memory evaluation

Evaluation uses binary masks, separately from the soft training objective. Within each fixed observation/query group, define

$$J_{ij} = \text{IoU}(\hat{Y}_i, Y_j), \quad d_i = J_{ii}, \quad (12)$$

$$w_i = \max_{j \neq i} J_{ij}, \quad \Delta_i = d_i - w_i. \quad (13)$$

Memory Fidelity is the reference-wise mean of $\mathbf{1}[d_i > w_i]$; ties, including all-zero rows, fail. With $\tau = 0.5$, define $a_i = \mathbf{1}[d_i > w_i \wedge d_i \geq \tau]$. CMSA is the fraction of identity pairs for which both references succeed:

$$\text{CMSA} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} a_i a_j. \quad (14)$$

Pairs are formed within each group; the primary set has two identities per group, so each group contributes one pair. Identity Error Rate is the reference-wise mean of $\mathbf{1}[w_i > d_i \wedge w_i \geq \tau]$. The mean identity margin averages Δ_i , while target mIoU averages d_i . Thus Fidelity can be correct despite poor localization, and IER is not its complement. Aggregation weights references equally for these reference-level metrics and pairs equally for CMSA.

The frozen Cycle025 runner loads base-w15 in BF16 and uses a fixed multi-round conversation containing SEG placeholders. It performs a direct forward pass, extracts helmet rows, and thresholds mask logits at zero; it does not generate an autonomous dialogue or select candidates using target IoU. Helmet targets are excluded from inference raster reads and supervision-mask placeholders are zero. Predictions are frozen before offline scoring. CMA and pinned SegLLM receive the same underlying observation/query and corresponding supplied miner identity information through their method-native memory interfaces, not identical internal memory encodings. The primary protocol is disjoint from documented training/evaluation image-byte registries; exact run-bound and ancestor exposure remain unknown.

2.5 Scope boundary

The evidence evaluates supplied-memory use under a fixed compound stressor. It does not establish autonomous memory writing, persistence or repair, a successful inference-time agent controller, safety guarantees, or isolated component superiority. Earlier Layer-2 controller/verifier attempts remain negative evidence and are not part of the validated method contribution.

3 Results and limitations

Evaluation contract. All degraded results below use this fixed photometric/noise/blur stressor. CMA and pinned SegLLM receive the same underlying observation/query and corresponding supplied miner identity information through method-native memory interfaces; internal memory encodings are not identical.

Table 1: Primary Cycle025 DOCUMENTED_PROTOCOL_DISJOINT_CONFIRMATION: 50 groups / 100 identity references per condition, with supplied identity memories and reconstructed pseudo targets. The comparison is system-level; exact run-bound w15 and ancestor exposure is unknown. Counts and rates are shown for CMSA, Memory Fidelity and IER; margins are unscaled. The set is disjoint from documented training/evaluation image-byte registries, not proven exact training-unseen.

Method	Condition	mIoU (%)	CMSA	Fidelity	IER	Mean margin
CMA base-w15	clean	88.28	46/50 (92%)	97/100 (97%)	1/100 (1%)	0.873944
CMA base-w15	target15_b	67.53	29/50 (58%)	84/100 (84%)	3/100 (3%)	0.646666
SegLLM pinned	clean	42.71	2/50 (4%)	49/100 (49%)	42/100 (42%)	0.049034
SegLLM pinned	target15_b	36.44	1/50 (2%)	46/100 (46%)	38/100 (38%)	0.031818

Counterfactual identity-memory evaluation. We hold the observation and semantic query fixed while switching the supplied miner identity memory, which changes the intended helmet target. Table 1 reports the primary 50-group documented-protocol-disjoint confirmation. CMA achieves 92% CMSA on clean observations and 58% under `target15_b`, compared with 4% and 2% for pinned SegLLM. Under degradation, Memory Fidelity is 84% versus 46%, and Identity Error Rate is 3% versus 38%. These results support stronger supplied-memory use on this evaluated set. Figure 2 illustrates the controlled identity switch and a retained degradation failure using previously frozen cases.

Robustness under compound degradation. CMA attains 67.53% mIoU under the fixed compound stressor, compared with 36.44% for SegLLM, a paired gap of 31.08 percentage points on identical groups. The supported claim is higher absolute degraded performance and retained identity fidelity: CMA still fails strict degraded CMSA on 21 of 50 groups. The overlap-affected historical and development replications are reported separately in Table 2 and are not pooled with the primary set.

Comparison scope. SegLLM is a pinned released historical-memory comparator receiving the same observations and supplied miner identities through its native interface. The comparison is system-level: training exposure and domain adaptation differ, and SegLLM’s lower clean performance limits architecture-only attribution. The counterfactual intervention measures identity-dependent behavior under common inputs; it does not isolate the contribution of an individual CMA component.

Limitations. The primary set was selected before inference and is disjoint by source-image SHA256 from reconstructed documented final-stage training and historical evaluation registries. Exact run-bound w15 and ancestor/pretraining exposure remain unknown, so this is not an exact training-unseen test. Memories are externally supplied and targets are reconstructed pseudo labels; 23 groups retain historical review-level episode QC and 27 combine accepted-high pairs. Structural validity does not replace independent annotation. CMA’s clean-

to-degraded mIoU decline is 20.75 points, versus 6.26 for SegLLM; we do not claim lower degradation sensitivity. The evidence does not establish autonomous memory writing, persistence or repair, broad corruption generalization, or safety guarantees. Earlier Layer-2 verifier/controller attempts did not meet their stated requirements and remain negative evidence, not a successful agent contribution.

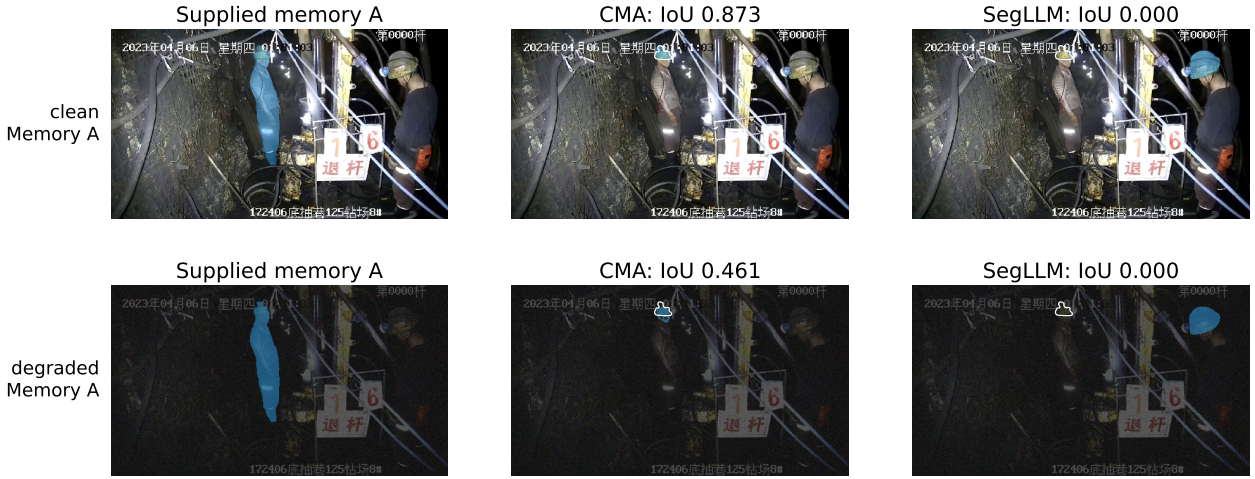
(a) Identity switch: same degraded observation, memory A vs B

cf_778571dc8020b5df



(b) Degradation limitation: fixed memory A, clean vs degraded

acceptedpair_b914b3d4a14a9083



White outline: saved pseudo target. Blue: identity A; orange: identity B.
Full frames; display-only mask overlays. IoU copied from frozen records.

Figure 2: Supplied identity-memory use and its degradation limitation. (a) The same degraded observation/query with memories A and B in the frozen success case. (b) Fixed memory A on clean and degraded observations in the frozen limitation case; memory B remains in the supplementary bank. Columns show memory, CMA and SegLLM. Blue/orange identify A/B; white contours are saved helmet pseudo-targets. IoUs are copied from frozen records. Full frames use display-only overlays without brightness or mask changes. The success case retains review-level episode QC; the limitation case combines accepted-high pairs. No new case selection or inference was performed.

Table 2: Separate overlap-affected replications: each block contains 50 groups and 100 identity references per condition. The reconstructed training protocol overlaps 37/50 Cycle022 images and 50/50 Cycle018 images; their source holdout was historically evaluated. Cycle018 also served development. These are not independent training-unseen tests. Supplied memories, reconstructed pseudo targets and system-level comparison limitations apply. No pooling is performed.

Method	Condition	mIoU (%)	CMSA	Fidelity	IER	Mean margin
<i>Cycle022: historical replication; 37/50 documented overlap</i>						
CMA base-w15	clean	90.87	47/50 (94%)	98/100 (98%)	2/100 (2%)	0.892993
CMA base-w15	target15_b	68.02	30/50 (60%)	87/100 (87%)	2/100 (2%)	0.657803
SegLLM pinned	clean	37.29	3/50 (6%)	43/100 (43%)	38/100 (38%)	0.041568
SegLLM pinned	target15_b	27.52	1/50 (2%)	38/100 (38%)	30/100 (30%)	0.016394
<i>Cycle018: development replication; 50/50 documented overlap</i>						
CMA base-w15	clean	91.97	47/50 (94%)	99/100 (99%)	0/100 (0%)	0.919718
CMA base-w15	target15_b	69.11	29/50 (58%)	89/100 (89%)	3/100 (3%)	0.662931
SegLLM pinned	clean	41.57	2/50 (4%)	47/100 (47%)	41/100 (41%)	0.047717
SegLLM pinned	target15_b	33.14	0/50 (0%)	42/100 (42%)	39/100 (39%)	-0.027858